

CONFIDENTIAL — ATTORNEY WORK PRODUCT

NexCort iQ / Apps on Demand LLC

Patent & Trademark Filing Guide

Explicit patentable processes and methods for each candidate
(PAT-01 through PAT-12) plus trademark filing directions

Prepared from: Competitive Intelligence Analysis (Sept 25, 2026)
and the NexCort iQ production codebase

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Engineering cross-check: September 25, 2026

Not legal advice. *This document is a technical enablement brief for patent counsel and trademark counsel. All claim language must be reviewed and rewritten by a registered patent attorney before filing.*

1. Purpose and how to use this guide

This guide maps each of the twelve NexCort iQ patent candidates to (a) what is already implemented in the product, (b) the specific **process / method** that is the inventive subject matter, (c) why that combination is more patentable than the underlying commodity features (transcription, translation, video), and (d) filing urgency. A separate section lists trademark marks, classes, and brand hygiene rules.

Patents protect **how** a technical result is achieved (processes, machines, manufactures, compositions — typically claimed as methods and systems). Trademarks protect **source identifiers** (names, logos, slogans) used in commerce. Do not confuse the two.

2. Executive filing priorities

Priority	Action	Window
P0	Provisionals: PAT-03 (RCS), PAT-01 (metered authz), PAT-02 (CAD write-back gate), PAT-04 (permission propagation)	~30 days
P0	USPTO word marks: NexCort iQ / NexCort; NexiQ Vision / NexiQ	~14 days
P1	Provisionals: PAT-05, PAT-06, PAT-07	~90 days
P1	Differentiated claims: PAT-09, PAT-11, PAT-12; careful draft of PAT-10 vs Axon Vision	Before public deep-dive demos
Skip standalone	Do not patent transcription, translation, or generic AI call handling alone	N/A

3. What makes software methods patentable (practical framing for counsel)

US patent eligibility for software is contested. Stronger applications emphasize a **specific technical process** that improves computer/system operation or life-safety workflow integrity—not an abstract idea of “using AI on 911 calls.” For each candidate below, claim the **ordered combination of steps**, the **fail-closed / gated machine behavior**, and the **data structures / state machines** that enforce safety—citing enablement from the shipped codebase.

- **Method claims:** numbered steps performed by processors / network systems.
- **System claims:** agency-scoped servers, authorization layer, CAD adapters, mobile clients, sensors.
- **Avoid:** claiming “AI transcription” or “translation” as the invention.
- **Prefer:** claiming how authorization, metering, gating, geofence confirmation, and audit interact as one machine.

PAT-01: Composable Feature-Level API Authorization with Atomic Usage Metering

Filing urgency: CRITICAL — file provisional now

Competitive verdict: STRONG CLEAR (competitive white space)

Code anchors: packages/shared/src/monetization/entitlements.ts; packages/shared/src/billing/addon-catalog.ts; apps/api addon/auth paths; tenant entitlements repository

Problem addressed

Public-safety AI vendors sell bundled enterprise plans. Agencies cannot activate a single capability (e.g., translation or RCS) with metered, enforceable authorization at the API boundary.

Patentable process / method (claim this ordered combination)

1. Receive an authenticated API or console request carrying a verified agency identity and role claims.
2. Resolve the agency's monetization plan and per-feature entitlement map (addon keys / feature flags) from an authorization store scoped by agencyId.
3. Atomically decide allow/deny for the specific feature key before executing the AI or workflow operation.
4. On allow, increment or reserve a usage meter (or seat/quota counter) in the same authorization transaction path—or immediately thereafter under an idempotent meter key—so usage cannot drift from authorization.
5. On deny, return a structured denial without performing the side-effecting operation; emit an audit event.
6. Propagate entitlement changes to subsequent requests without requiring the client to rotate API credentials (links to PAT-04).

Why this is patentable (vs commodity features)

The inventiveness is not “having an API” or “billing.” It is the **ordered combination** of feature-level authorization, agency-scoped entitlement resolution, and atomic metering at the authorization layer for public-safety AI capabilities sold à-la-carte. Competitors expose bundled products or inbound data APIs; they do not meter outbound AI features this way.

Claim drafting directions for counsel

- Independent method: authorize → meter → execute / deny.
- Dependent: specific feature keys (cad.writeback_*, rcs.module, call_assist.module, translation).
- Dependent: fail-closed when entitlement store unavailable.
- System claim: authorization service + meter store + AI/workflow services.

PAT-02: AI CAD Write-Back with Supervisor Approval Gating (Fail-Closed)

Filing urgency: CRITICAL — file provisional now

Competitive verdict: STRONG CLEAR

Code anchors: apps/web/lib/rapid-cortex/cad/cad-writeback-gate.ts; CAD write-back approval queue UI; API env CAD_WRITEBACK_ENABLED default false; audit on blocked write-back; addon catalog cad.writeback_*

Problem addressed

AI systems that write into CAD either write directly (unsafe) or never write (limited value). Agencies need enrichment that cannot reach the system of record without human approval and explicit legal/ops enablement.

Patentable process / method (claim this ordered combination)

1. Generate a proposed CAD write-back payload from AI or dispatcher-assist analysis (incident fields, narrative fragments, codes).

2. Gate 1 (environment / legal): refuse write-back unless an explicit environment enablement flag is true (default false / fail-closed).
3. Gate 2 (role): require an authorized human role (e.g., dispatcher approval and/or supervisor queue) before any CAD adapter invoke.
4. Persist the proposal in an approval queue with agencyId scope, proposer identity, and immutable proposal content hash.
5. On approval, invoke the CAD adapter write; on denial or missing enablement, block the write and record an audit event (blocked write-back).
6. Never auto-promote AI output to CAD without traversing both gates.

Why this is patentable (vs commodity features)

Prior art includes CAD read-only feeds and direct integrations. The patentable method is the **fail-closed dual gate** (environment/legal + human approval) applied specifically to AI-derived CAD mutations, with audit on blocked paths.

Claim drafting directions for counsel

- Independent: dual-gate AI→CAD write method.
- Dependent: signed addendum / contract flag as part of Gate 1.
- Dependent: multi-vendor CAD adapters behind the same gate.
- Avoid claiming “AI suggests CAD fields” alone.

PAT-03: Response Continuity System (Silent Monitor + Geofence + Audio Sentinel + Closure Gate)

Filing urgency: CRITICAL — strongest product claim; file now

Competitive verdict: STRONG CLEAR — highest IP value per competitive analysis

Code anchors: packages/shared/src/rcs/types.ts & schemas.ts — states
 MONITORING→...→CLOSED/OVERRIDE_CLOSED; arrivalRadiusMeters; audioStatus; supervisorAck;
 closureOverride; addon rcs.module

Problem addressed

Life-safety calls can lose continuity after initial dispatch. Point solutions exist for call monitoring, fleet geofence, or audio analytics—but not as one enforced continuity machine for emergency response.

Patentable process / method (claim this ordered combination)

1. Create a continuity call record bound to agencyId with an initial state of MONITORING (silent monitor queue).
2. Track responding units' positions against a call geofence (arrivalRadiusMeters); transition to UNIT_ARRIVED only when a unit is confirmed on-scene.
3. Run an audio sentinel state machine (SILENT / LISTENING / ALERT / CONFIRMED_SAFE / CONFIRMED_DANGER) that can raise escalation levels.
4. On escalation, require supervisor acknowledgement (supervisorAck) before de-escalation or closure pathways proceed.
5. Enforce a closure gate: the call cannot reach CLOSED unless continuity conditions are satisfied; alternatively allow OVERRIDE_CLOSED only with a recorded supervisor override (byUserId, badge, reason, timestamp).
6. Emit audit/WebSocket events for arrival, escalation, handoff, and closure for the agency console.

Why this is patentable (vs commodity features)

Novelty is the **unified state machine** combining silent monitoring, geofence arrival confirmation, audio sentinel escalation, and supervisor-gated closure with override audit—for emergency response continuity. Competitors have pieces; none combine this life-safety gate set.

Claim drafting directions for counsel

- Independent: four-module continuity method with terminal states CLOSED / OVERRIDE_CLOSED.
 - Dependent: soft handoff request/accept between dispatchers.
 - Dependent: AI summary generation attached to continuity record without bypassing closure gate.
 - System: mobile unit clients + agency RCS service + supervisor console.
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PAT-04: Real-Time API Permission Propagation Without Credential Rotation

Filing urgency: CRITICAL — file with PAT-01

Competitive verdict: STRONG CLEAR (claims must match shipped mechanism)

Code anchors: apps/api/src/repositories/tenantEntitlementsRepository.ts; Cognito custom claims; feature-flag / entitlement resolution on request path

Problem addressed

Changing which AI features an agency may use typically forces API key rotation or redeployed credentials—slow and operationally risky for 911 centers.

Patentable process / method (claim this ordered combination)

1. Authenticate the caller with a durable credential (e.g., Cognito JWT / API key) that identifies agency and principal but does not embed the full mutable feature set as the sole source of truth.
2. On each privileged request (or on a short-lived cache TTL), fetch the current entitlement map for agencyId from a server-side store.
3. Authorize the specific feature against the live map; deny if revoked—even if the client credential is unchanged.
4. When an admin enables/disables an addon, update the entitlement store; subsequent requests observe the change without issuing new client secrets.
5. Optionally push invalidation events to connected consoles so UI capability sets update in real time.

Why this is patentable (vs commodity features)

The inventive concept is **separating durable identity credentials from mutable feature entitlements** and enforcing live server-side resolution so permission changes propagate without credential rotation—in a multi-tenant public-safety AI platform.

Claim drafting directions for counsel

- Independent: live entitlement resolution per request without key rotation.
 - Dependent: combination with PAT-01 atomic metering.
 - Counsel must map claims to exact cache/TTL/invalidation implementation to avoid abstract “permissions” rejection.
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PAT-05: Browser→Cloud→HMAC→On-Prem PTZ Gateway (Camera IPs Never Exposed)

Filing urgency: HIGH — ~90 days

Competitive verdict: CLEAR in public-safety domain

Code anchors: Rapid Vision / camera registry / producer secrets / KVS paths; HMAC-gated on-prem bridge patterns

Problem addressed

Dispatch consoles need PTZ/camera control without exposing on-prem camera network addresses to the public internet or browser.

Patentable process / method (claim this ordered combination)

1. Browser issues a PTZ/control intent to a cloud control plane after user authorization for the incident/agency.
2. Cloud validates role and incident scope; never returns raw camera LAN IPs to the browser.
3. Cloud constructs a signed command (HMAC or equivalent) addressed to an on-prem gateway identity.
4. On-prem gateway verifies signature, maps logical camera ID → private IP, executes PTZ/control on the local network.
5. Telemetry/acks return through the cloud; browser receives status only.

Why this is patentable (vs commodity features)

Domain-specific method that keeps camera addressing private while enabling authorized remote PTZ through a signed cloud-to-edge hop—distinct from generic VMS cloud portals.

Claim drafting directions for counsel

- Independent: signed cloud→edge PTZ with IP non-disclosure to clients.
 - Dependent: incident-scoped authorization, time-limited command tokens.
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PAT-06: Citizen Camera Consent with Hashed Single-Use Token + IP Rate Limit

Filing urgency: HIGH — ~90 days

Competitive verdict: CLEAR — no public-safety analogue cited

Code anchors: Nest/Ring consent flows; consent HMAC secrets; camera provider OAuth state tables

Problem addressed

Emergency access to citizen/facility cameras requires informed consent without reusable tokens that can be stolen or replayed.

Patentable process / method (claim this ordered combination)

1. Dispatcher initiates a consent request for cameras near an incident.
2. System issues a single-use consent token to the device owner channel (SMS/app).
3. Store only a one-way hash of the token (e.g., bcrypt/argon) server-side; never store the raw token at rest.
4. On redemption, verify hash, mark token consumed, and apply IP/request rate limits to block brute force.
5. Grant time-bounded viewing only after successful redemption; audit the consent event.

Why this is patentable (vs commodity features)

Combination of emergency-workflow consent issuance, hashed single-use redemption, and rate limiting for camera access—not merely OAuth login.

Claim drafting directions for counsel

- Independent: hashed single-use consent redemption for emergency camera access.
 - Dependent: auto-expiry, agency-scoped audit, provider-specific adapters (Ring/Nest/Wyze).
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PAT-07: QR/NFC Venue (Campus/Transit) Reporting with Automatic Physical Location Extraction

Filing urgency: HIGH — ~90 days

Competitive verdict: CLEAR — venue/campus white space

Code anchors: packages/shared/src/qr-nfc/*; campus/venue/transit QR admin; Field app NFC write

Problem addressed

Guests/students report issues without knowing formal addresses; staff need automatic mapping to building/zone/section.

Patentable process / method (claim this ordered combination)

1. Provision a QR/NFC tag bound to a structured location record (campus/venue/transit zone IDs, human label, coordinates as applicable).
2. On scan/tap, open an intake surface that resolves tag ID → location context server-side (not typed by the reporter).
3. Create an incident/report pre-populated with extracted physical location metadata and vertical-appropriate routing.
4. Enforce that PSAP roles cannot manage these QR inventories (product RBAC), preserving vertical separation.

Why this is patentable (vs commodity features)

Automatic physical-context extraction from QR/NFC into a multi-vertical public-safety intake workflow—not generic QR forms.

Claim drafting directions for counsel

- Independent: tag→location resolution→incident creation method.
- Dependent: NFC write via Field app; multi-campus tenant filtering.

PAT-08: Vertically-Agnostic Runtime Agency Config from DynamoDB (Zero Hardcoding)

Filing urgency: MEDIUM — provisional optional; consider trade secret

Competitive verdict: MEDIUM CLEAR — architectural

Code anchors: agency type / vertical in tenancy schemas; role routing; runtime feature config

Problem addressed

Adding campus/venue/hospital/transit usually forks codebases. A single platform must reconfigure behavior per agency without redeploying vertical-specific binaries.

Patentable process / method (claim this ordered combination)

1. Store agency type/vertical and feature configuration in a tenant record (DynamoDB).
2. At request/console bootstrap, load config and select navigation, protocols, and integrations dynamically.
3. Route users to vertical homes (PSAP vs venue vs campus vs transit vs call-assist) based on normalized role tokens preserved before legacy migration.
4. Avoid compile-time hardcoding of vertical behavior into separate products.

Why this is patentable (vs commodity features)

Thinner patent; strength is in concrete runtime resolution + role preservation algorithms. May be better as trade secret + copyright if claims feel abstract.

Claim drafting directions for counsel

- Focus on normalizeSessionRole before legacy migration to prevent vertical collapse.
 - Pair with PAT-01 entitlements for stronger system claims.
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PAT-09: One-Bot-Per-Agency NLU Isolation with Hardcoded TRANSFER_911 Path

Filing urgency: HIGH — differentiate vs Axon/RapidSOS non-emergency AI

Competitive verdict: MEDIUM CLEAR — safety path novel

Code anchors: Call Assist / Lex stacks; call_assist.* addons; voice-config per agency

Problem addressed

Shared multi-tenant NLU bots risk cross-agency leakage and unsafe handling of true emergencies in “non-emergency” lines.

Patentable process / method (claim this ordered combination)

1. Provision an isolated NLU/bot configuration per agency (no shared intent model that can leak utterances across tenants).
2. Classify inbound non-emergency dialogue with agency taxonomy.
3. If emergency indicators fire, take a hardcoded TRANSFER_911 (or equivalent) path that cannot be disabled by agency prompt editing.
4. Log the transfer decision with audit; never allow LLM prompt overrides to remove the emergency escape hatch.

Why this is patentable (vs commodity features)

Isolation + **non-overridable emergency transfer** in a non-emergency AI call system—distinct from generic callbots.

Claim drafting directions for counsel

- Independent: per-agency bot isolation with immutable emergency transfer path.
 - Dependent: TTY accommodation, external routing addons.
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PAT-10: AI Scene Observation Text Stream to Dispatcher Console

Filing urgency: HIGH — draft after Axon Vision prior-art memo

Competitive verdict: MEDIUM — Axon Vision partial overlap

Code anchors: packages/shared rapid-vision types; scene-intel; WebSocket events rapid-vision.observation.* / transcript.segment

Problem addressed

Dispatchers need continuous, human-readable scene observations from live video without replacing the VMS.

Patentable process / method (claim this ordered combination)

1. Establish an incident-scoped vision session with agency authorization (canRequestVisionAccess).
2. Sample frames / run detectors on a schedule; generate observation text objects.
3. Push observations to the dispatcher console via typed WebSocket envelopes ({ type, data }) with GSI fallback for connections.
4. Allow verify/reject of observations with audit; do not auto-write to CAD (compose with PAT-02 if enrichment suggested).

Why this is patentable (vs commodity features)

Only if claims emphasize **incident-scoped session + observation verify workflow + envelope protocol**—not “AI looks at video.” Differentiate from Axon Vision explicitly.

Claim drafting directions for counsel

- Avoid broad “detect objects in video” claims.
 - Claim observation lifecycle: created→verified/rejected with dispatcher UX constraints.
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PAT-11: SMS Magic Link → PWA Field Brief Without App Installation

Filing urgency: HIGH — method claims

Competitive verdict: MEDIUM — Axon requires app; method novel

Code anchors: Field brief / Pinpoint SMS paths; mobile PWA surfaces

Problem addressed

Field units need incident briefs quickly without mandating a native store-installed app.

Patentable process / method (claim this ordered combination)

1. Authorize a brief for a unit/user and incident.
2. Send an SMS containing a time-limited magic link (Pinpoint/SMS provider).
3. On open, authenticate via the link token and render a progressive web app field brief in the mobile browser.
4. Expire or revoke the link; audit access; do not require App Store/Play install for the brief path.

Why this is patentable (vs commodity features)

End-to-end method of SMS magic-link delivery to an install-free PWA brief for public-safety field use.

Claim drafting directions for counsel

- Independent: SMS token → PWA brief method.
 - Dependent: offline cache limits, CJIS-aware session timeout.
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PAT-12: Multi-Axis Non-Emergency Taxonomy + LLM + Hardcoded 911 Override

Filing urgency: HIGH — combination claims only

Competitive verdict: MEDIUM — combination novel; parts exist

Code anchors: Triage / Call Assist taxonomy; LLM assist; emergency override paths

Problem addressed

LLM classification alone is unsafe for public-safety telephony; pure taxonomies lack flexibility.

Patentable process / method (claim this ordered combination)

1. Apply a multi-axis taxonomy (intent, urgency, location certainty, vulnerability, etc.) to an inbound non-emergency interaction.
2. Use an LLM to assist labeling / summarization within taxonomy constraints.
3. Independently evaluate hardcoded emergency override rules; if triggered, force 911/PSAP transfer regardless of LLM output.
4. Persist both taxonomy labels and override decisions for QA/audit.

Why this is patentable (vs commodity features)

Patent the **combination** of multi-axis taxonomy + LLM assist + non-bypassable 911 override—not LLM classification alone.

Claim drafting directions for counsel

- Independent: taxonomy+LLM+hardcoded override pipeline.
 - Explicitly disclaim monopoly on transcription/translation.
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4. Do NOT file as standalone inventions

- Real-time 911 call transcription — Axon Prepared, Motorola Smart Transcription, RapidSOS Harmony.
- Real-time translation for 911 — Axon/Carbyne and others.
- Generic non-emergency AI call handling — market crowded; protect via PAT-09 / PAT-12 combinations and PAT-01 licensing.

5. Trademark filing directions

Trademarks identify source of goods/services. File in the United States (USPTO) with counsel; consider Madrid Protocol later for Canada/EU if expanding.

5.1 File immediately (word marks)

Mark	Use in product	Notes / classes (starting point)
NexCort iQ / NEXCORT IQ	Primary platform brand (site.ts, invoices, packets)	Classes 9, 38, 42, 45 — confirm with counsel
NexCort	House mark / family	Same class set; blocks lookalikes
Apps on Demand	Legal entity / operator	Class 42 (+ others as used in commerce)

5.2 File as product marks (use or ITU)

Mark	Priority	Direction
NexiQ Vision	High	Already used with ™ in marketing — file before wider campaigns
NexiQ	High	Product-line house mark (Vision, Field, sales intel). Pick consistent styling vs NexCort iQ
Call Assist	Medium	Expect descriptiveness scrutiny; secondary meaning / logo lockup may help
Guest Assist	Medium	Venue guest intake brand
Response Continuity System	Medium	Prefer full phrase; “RCS” alone is weak
Intelligence at the speed of response	Medium	Slogan mark if used in advertising/commerce consistently

5.3 Trademark carefully / usually skip

- **Rapid Cortex** — former name; keep for transition copy, do not lead new brand filings unless still selling under it.
- **Loadout** — appears in competitive strategy docs but not as a billing product key in shared code; only file if you will brand the marketplace that way in commerce.
- **RCS** alone — too generic across industries.
- **Pinpoint / RC Lite** — crowded or descriptive; full clearance search first.

5.4 Brand hygiene rule (strengthens TM)

- **NexCort iQ** = company / platform.
- **NexiQ** ____ = product lines (Vision, Field, sales intel).
- Use ™ before registration; ® only after federal registration issues.
- Record first-use dates (screenshots, invoices, domain launches) for each mark.

6. Provisional patent package checklist (per candidate)

- Title + field of invention (public-safety AI / emergency communications software).
- Background: problem with bundled vendors and unsafe AI→CAD writes / lost call continuity.
- Summary of invention: the ordered method steps from this guide.
- Detailed description: screenshots + sequence diagrams + references to code modules listed herein.
- At least one independent method claim and one system claim (counsel-drafted).
- Inventor declarations (engineering contributors who conceived claimed subject matter).
- Do not publicly disclose claim details beyond NDAs until provisional is filed (or accept grace-period risk).

7. Sources

- NexCortiQ Competitive Analysis (Confidential), Jeffrey W. Coleman Jr., Sept 25, 2026 — 8 competitors × 12 patent candidates.
- NexCort iQ monorepo: packages/shared (rcs, monetization, qr-nfc, billing), apps/api, apps/web CAD write-back gate, Rapid Vision.

8. Sign-off

Role	Name	Date
Prepared (technical brief)	Engineering assist / CEO materials	2026-09-25
Patent counsel review	_TBD_	
Trademark counsel review	_TBD_	

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